

2023



Daiso Built Data Analysis Tools Using AWS, Helping All Employees to Make Data-Driven Business Decisions

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Nearly 16 million yen saved

in operating costs for BI tools

200 user base

of Amazon QuickSight

76,000 items

managed by the centralized POS data



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Overview

[Daiso Industries](#) (Daiso) is a Japan-based company that operates the Daiso 100-yen store. The company migrated to the cloud in 2013 and has adopted Amazon Web Services (AWS). In 2018, it built a



point-of-sale (POS) data processing system and introduced new tools for data analysis. Now, the company provides dashboards to all employees for purchasing and developing products and other applications.



Opportunity | Establishing a Centralized POS Data Processing System to Handle Large Volumes of Product Data

Daiso develops original products such as stationery, cosmetics, tableware, and lifestyle goods, and it sells these products under brands such as DAISO, THREEPPY, and Standard Products. The company, which sells 76,000 products in nearly 6,000 stores, decided to migrate to the cloud in 2013 to improve its efficiency in system development and chose to build its ordering system on AWS. In 2014, Daiso built an automated ordering system using [Amazon Redshift](#). The company's next step was to make the system serverless. The company built part of the merchandise management system using [AWS Lambda](#). Then, in 2018, Daiso built a centralized POS data processing system using AWS Lambda and other tools to process merchandise sales data from each store.

The new centralized POS data processing system helps the company to scale compute resources even when a large amount of data is concentrated, providing stable processing without being affected by peak periods. In addition, the previous environment could store data for only 2 years, but the use of [Amazon Simple Storage Service](#) (Amazon S3) has provided long-term storage. The ability to analyze supply and demand in conjunction with data from global locations has had a significant business impact on the company as it drives global expansion. Currently, Daiso has expanded its POS data collection to the United States, Singapore, Malaysia, India, and other countries, and it plans on expanding the collection area sequentially in the future.

Solution | Developing Departmental Dashboards Using Amazon QuickSight

The next step of Daiso's serverless initiative was using a business intelligence (BI) tool for data visualization. The company's previous tool couldn't retain the data in the database for the necessary length of time for analysis. Therefore, the company adopted [Amazon QuickSight](#) which was launched for the Tokyo region in 2018. "The key factor was the significant increase in the retention period of store inventory data from 1 day to 2 years. This long-term data retention helps us to see trends in sales, sales volume, store inventory, and warehouse inventory on an item-by-item basis," says Takaya Kanzaki, manager of the system planning section, information systems department, corporate planning division of Daiso. "Another attractive feature was the per-session pay-as-you-go billing system. Amazon QuickSight has two types of licenses: Author, for dashboard creators, and Reader, for viewers. Reader licenses are not charged if not used."

The first Amazon QuickSight implementation was completed in 2 months and started with 35 buyers in the merchandise department at Daiso headquarters. The running cost of the BI tool was reduced by more than 1.3 million yen per month, or nearly 16 million yen per year, including server configuration fees.

In 2020, the information systems department decided to build a departmental dashboard with the goal of having everyone at headquarters use the data. "The idea was that visualizing data in the form of a dashboard for our employees would encourage use of Amazon QuickSight," says Takaya Kanzaki. To build the dashboard, the team conducted interviews with key personnel in each department, including management, store operations, and merchandise, to confirm the information they wanted to visualize. In January 2022, the first management dashboard was released, followed by a series of departmental dashboards.

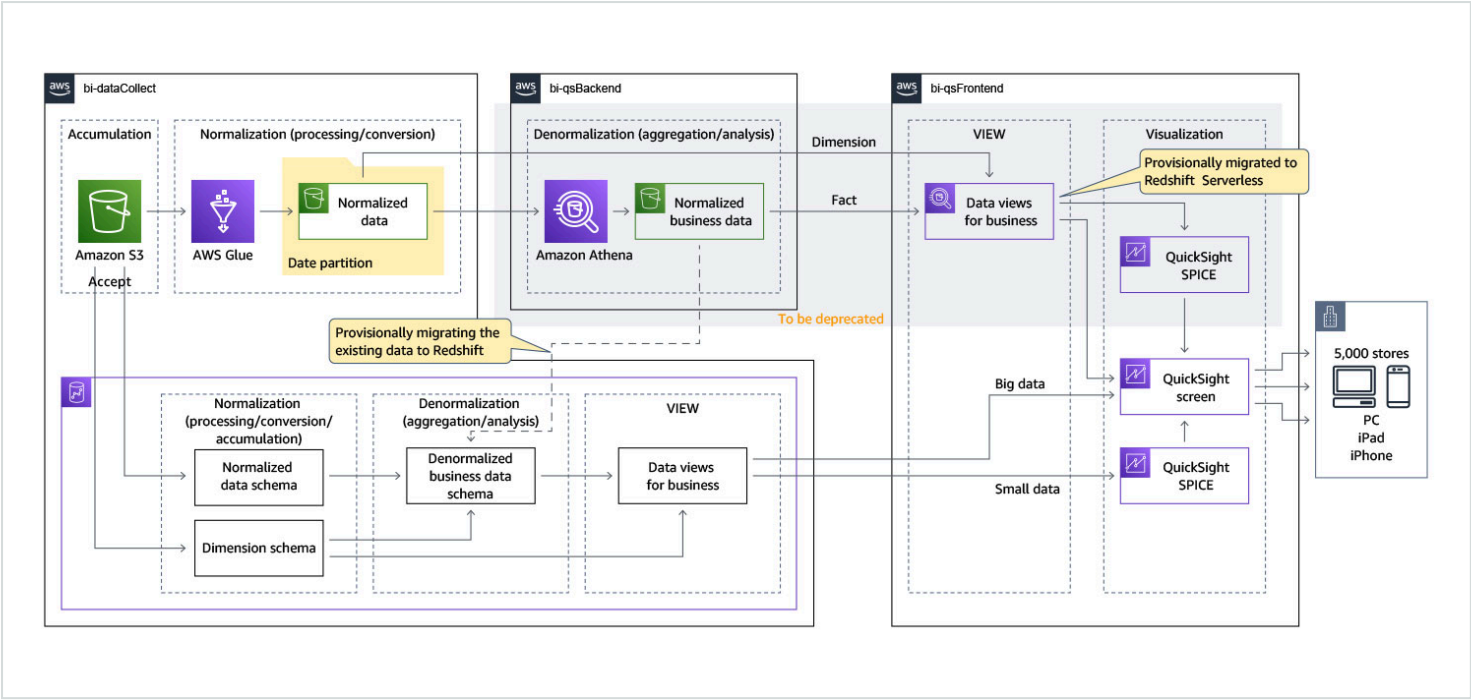
"We held discussions as we built," says Kanzaki. "Management had a variety of requests, including the need to see cost information, actual versus budget ratios, year-over-year ratios, and long-term trends of those metrics, which are necessary for business decisions, as well as the need to see the previous day's data at 7:00 a.m. The store development department wanted to see new store trends and simple 10-day sales trends; the store operations department wanted to see sales comparisons before and after the COVID-19 pandemic as well as budget achievement status; and the merchandise department wanted to see sales by category and by region."

In April 2022, the dashboards were rolled out to the United States and Singapore, where local and store managers are now able to see sales trends and other information in approximately United States and 50 stores in Singapore.



As of December 2022, Daiso has released around 40 dashboards company-wide, and each department is using them. Releasing the dashboards has increased users' motivation because they are able to see the data they need. The Daiso team has received many requests about what indicators users want to see, and the team continually communicates with key users and updates the dashboards as needed.

Architecture



Outcome | Facilitating Company-Wide Data Sharing and Use

As of December 2022, Amazon QuickSight is available to nearly all employees. “According to the most recent survey, almost all the employees are actively using the system,” says Kanzaki. “In addition to the main user groups of the system—such as management, the merchandise department, and the store development division—the logistics department is trying to optimize the supply chain while visualizing the inventory status of domestic and overseas warehouses in conjunction with the warehouse management system. The store management division visualizes sales trends by store brand, the usage ratio of electronic payments, and so on.”

Going forward, Daiso anticipates expanding Amazon QuickSight to stores across Japan and supporting store managers' decision-making according to store characteristics in order to democratize data use for all employees and drive data-driven management. With increase of access to the company's BI tool, Daiso plans to migrate the tool's backend from [Amazon Athena](#) to Amazon Redshift because Amazon Redshift is more cost-effective. Daiso is also envisioning predictions using AWS machine learning services and near-real-time data collaboration in the future.

“In terms of marketing, we are considering the use of machine learning for analyses that are difficult to perform manually, such as on the synergistic effects of products bought in combination or store brands,” says Kanzaki. “As sourcing becomes more difficult on a global scale, accurate inventory movement, including that of e-commerce sites, can help optimize inventory and increase sales. We expect AWS to continue to provide the latest information and support, from Amazon QuickSight enhancements and user interface improvements to the machine learning space.”

About Daiso Industries

Daiso Industries runs more than 6,000 DAISO 100-yen stores in Japan and 26 other countries and regions worldwide. It carries nearly 76,000 items and develops approximately 1,200 new products every month. In recent years, the company has developed the 300-yen shop THREEPPY and a new business model, Standard Products. It expanded its sales channels by opening an e-commerce site in 2020.





Mr. Takaya Kanzaki

AWS Services Used

Amazon QuickSight

Amazon QuickSight powers data-driven organizations with unified business intelligence (BI) at hyperscale.

[Learn more »](#)

Amazon Athena

Amazon Athena is a serverless, interactive analytics service built on open-source frameworks, supporting open-table and file formats. Athena provides a simplified, flexible way to analyze petabytes of data where it lives.

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AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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